

Getayawkal Girma

Addis Ababa | Email | GitHub | LinkedIn | Website

Experience

Full Stack Developer (*Frappe / ERPNext, Next.js, PostgreSQL*)
Oct 2024 – Present

Habtech Solutions

- Led end-to-end design and delivery of a Construction Project Management System (CPMS) for Ethiopia's Ministry of Health — architecting a multi-module Frappe application with clearly separated concerns across backend services, REST APIs, and UI components.
- Currently building a full-stack research data platform in Next.js and PostgreSQL, featuring a learning management system (LMS), file management, secure download-request workflows with token-based link generation, and a dynamic form engine powered by SurveyJS.
- Implemented role-based access control (RBAC) using BetterAuth — restricting Google OAuth login to specific user roles while supporting credential-based login for others.
- Integrated ODK (Open Data Kit) for offline field data collection, building dashboards to surface submission statistics and enumerator activity from the ODK server.
- Hardened deliverables for production: Dockerized the application, deployed to a self-hosted server, and obfuscated/compiled the frontend bundle to protect proprietary source code.
- Contributed an open-source bug fix to *Frappe Insights*: identified, reproduced, and resolved a permission validation issue in child table access checks, following a structured debugging and patch workflow.
- Authored technical documentation for system modules to support handoff, onboarding, and long-term maintainability across a distributed team.
- Built and validated REST API endpoints with Postman collections, ensuring deterministic, reproducible behaviour across environments.

Full-Stack Developer (.NET Core, Angular)
Feb 2025 – Aug 2025

AAU

- Developed and shipped five IMIS modules (Research, Workflow, Notice, Memo, Delegation) for a foreign affairs management system — working across a large multi-file Angular + ASP.NET Core codebase with clearly defined service boundaries.
- Refactored a monolithic backend into modular, independently testable services; enforced separation of concerns and clean dependency injection patterns throughout.
- Implemented server-side pagination and optimized SQL queries, reducing load time by 45% and improving reproducibility of data-heavy analytical views.
- Generated structured PDF reports with QuestPDF, producing consistent, deterministic document output from raw structured data.

Backend Engineer / AI Tooling Contributor (*FastAPI, Python, LLM Training*) Turing (Contract)
Jan 2025 – Dec 2025

- Collaborated on LLM training projects focused on teaching AI agents to safely and correctly use real-world developer tools.
- Designed and implemented FastAPI-based backend services that simulate real APIs (Stripe, Google Drive, PayPal) inside controlled training environments.
- Built multi-API “gym” environments that allow AI agents to practice tool usage with realistic request/response flows and failure scenarios.

- Implemented strict validation, permissions, and safety constraints to prevent destructive actions during agent training.
- Modeled APIs using Pydantic schemas to enforce structured inputs, outputs, and error handling.
- Created task-oriented endpoints that mirror real developer workflows such as payments, file management, and account operations.
- Worked closely with prompt designers and evaluators to improve agent reliability, correctness, and tool-selection behavior.
- Wrote tests and sanity checks to ensure deterministic behavior of training environments across runs.

Intern (.NET Core, Angular)

AAU

Sep 2024 – Jan 2025

- Resolved 20+ integration bugs across a large-scale Angular + ASP.NET Core codebase; documented root causes and fixes to prevent regression.
- Built notification and delegation modules with clearly defined interfaces; implemented multilingual support (English/Amharic) via i18n.

Projects

Mezgeb – Offline Receipt Manager

APK / GitHub

Ionic, Angular, TypeScript, SQLite, Regex, GitHub Actions CI/CD

- Designed a modular architecture across 9 services and 72 source files (7.8K+ LOC) with strict dependency injection and clean separation of concerns — structured to support long-term maintainability and independent testability of each module.
- Built a deterministic parsing engine using 40+ multilingual regex patterns, achieving 98% accuracy across 4,000+ real bank receipts (SMS, PDF, XML formats); each pattern is independently verifiable against a fixed test corpus.
- Automated the full release pipeline with GitHub Actions CI/CD — from build to signed APK distribution — ensuring consistent, reproducible output on every push.
- Integrated a custom Capacitor native plugin with documented interfaces; produced inline code documentation and architectural notes to support onboarding and future contributors.

ZDish – Restaurant Discovery & Menu Management Platform

Live Demo

Next.js 16, React 19, TypeScript, Prisma ORM, PostgreSQL (Supabase), Docker, LLMs

- Architected a 4-layer clean architecture (Actions → Services → Repositories → DB) following Domain-Driven Design, with a multi-file Prisma ORM schema isolating four data domains — structured to mirror real-world enterprise codebases.
- Containerized the development environment with Docker, enabling fully reproducible local setup and consistent behaviour across machines without manual environment configuration.
- Implemented tag-based cache invalidation (`revalidateTag`) with deterministic rules per domain, ensuring predictable, testable cache state transitions across the application.
- Authored structured technical documentation covering architecture decisions, domain boundaries, and data flow — enabling new contributors to onboard and extend the system independently.

Find the Imposter – Social Deduction Game

Live Demo

React Native (Expo), TypeScript, SQLite, Appwrite, Python, LLMs

- Developed an automated Python ETL pipeline with deterministic deduplication logic and human-in-the-loop verification gates — ensuring only reviewed, validated content reaches production.

- Implemented an offline-first SQLite architecture with a testable content-selection algorithm; designed the sync mechanism to be idempotent and safe to replay without side effects.
- Built bilingual (English/Amharic) content pipelines with AI-powered translation and structured categorization, each stage independently executable and verifiable.

Education

Bachelor of Mechatronics

Addis Ababa Science and Technology University

Graduated: 2024

Relevant Coursework: Object-Oriented Programming in Python, Machine Learning, Computer Vision, Algorithms & Data Structures, Embedded Systems, Integrated Engineering Team Project

Technical Skills

Languages: Python, TypeScript, C#, SQL, JavaScript, Node.js

Frameworks: Next.js, FastAPI, Angular, Ionic, ASP.NET Core / .NET Core, React, React Native,

Databases: PostgreSQL, SQLite, MySQL, MariaDB, Redis

ORM & Validation: Prisma ORM, Pydantic, Zod

Tools & DevOps: Docker, Git, GitHub Actions (CI/CD), Nginx, Supabase, Vercel, Appwrite, Postman

Auth & Security: BetterAuth, OAuth 2.0, JWT, Role-Based Access Control (RBAC)

Practices: Clean Architecture, Domain-Driven Design (DDD), Modular Design, Dependency Injection, REST API Design, Server-Side Rendering (SSR), TDD, Deterministic Testing, Reproducible Builds, Technical Documentation, Open-Source Contribution